

Networking Module 4b

DNSSec & Routing

CREATING TOMORROW



Agenda

- **DNS**
- Routing
- Routing Algorithms
- Routing Protocols

DNS

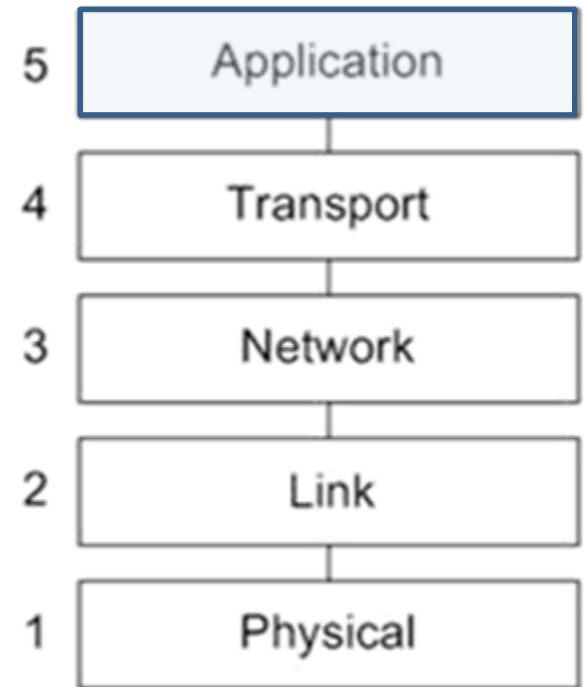
Reminder: DNS

DNS security risks:

- Use of DNS servers in DDOS attacks
- DNS hijacking

Possible solutions:

- DNS over HTTPS
- DNSSEC



TCP/IP Model

Figure 1

DNS name resolution

DNS services

- hostname to IP address translation
- host aliasing
- mail server aliasing
- load distribution

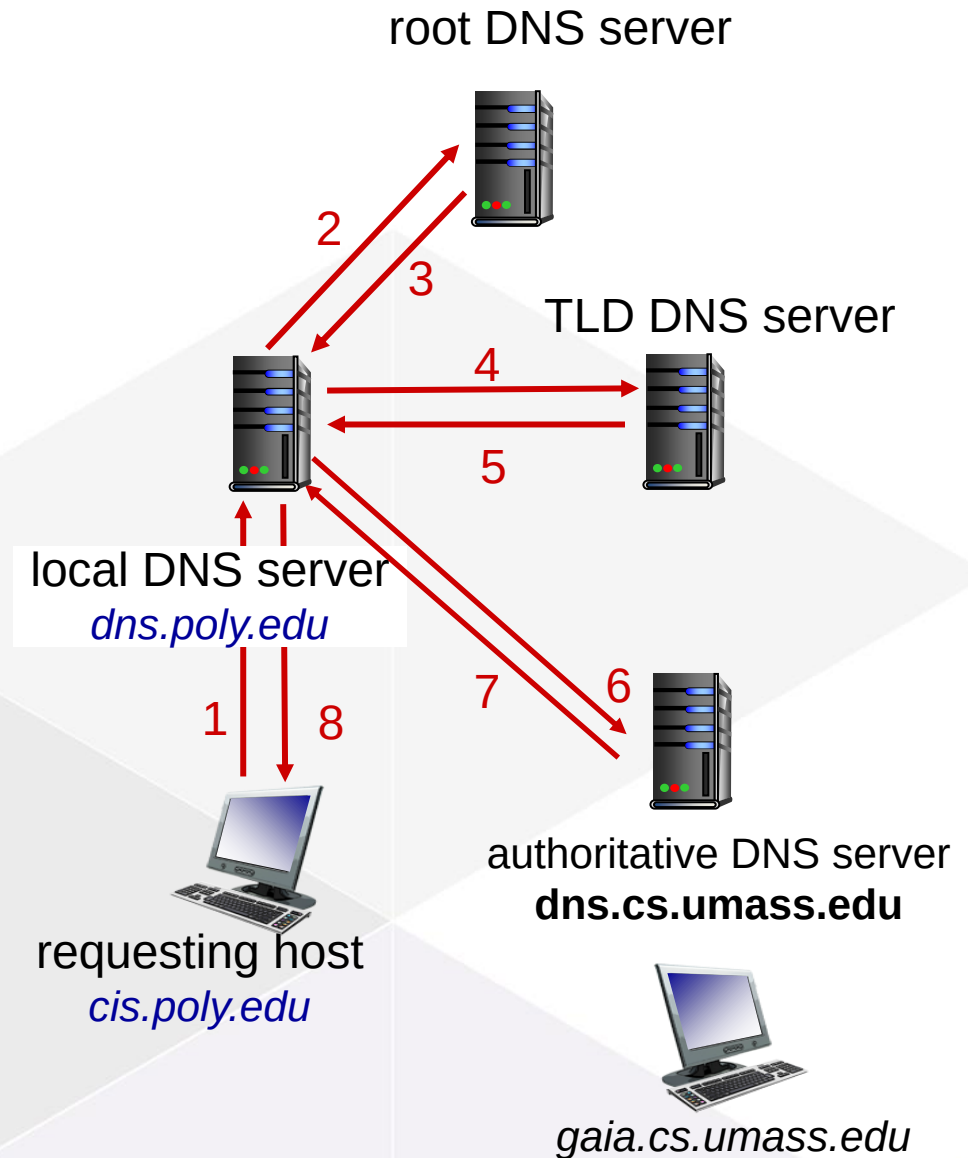


Figure 3

DNS in DDOS attacks

DNS servers can be used for DDOS amplification attacks

1. Attacker sends small DNS requests with fake source IP, requesting max info about zone

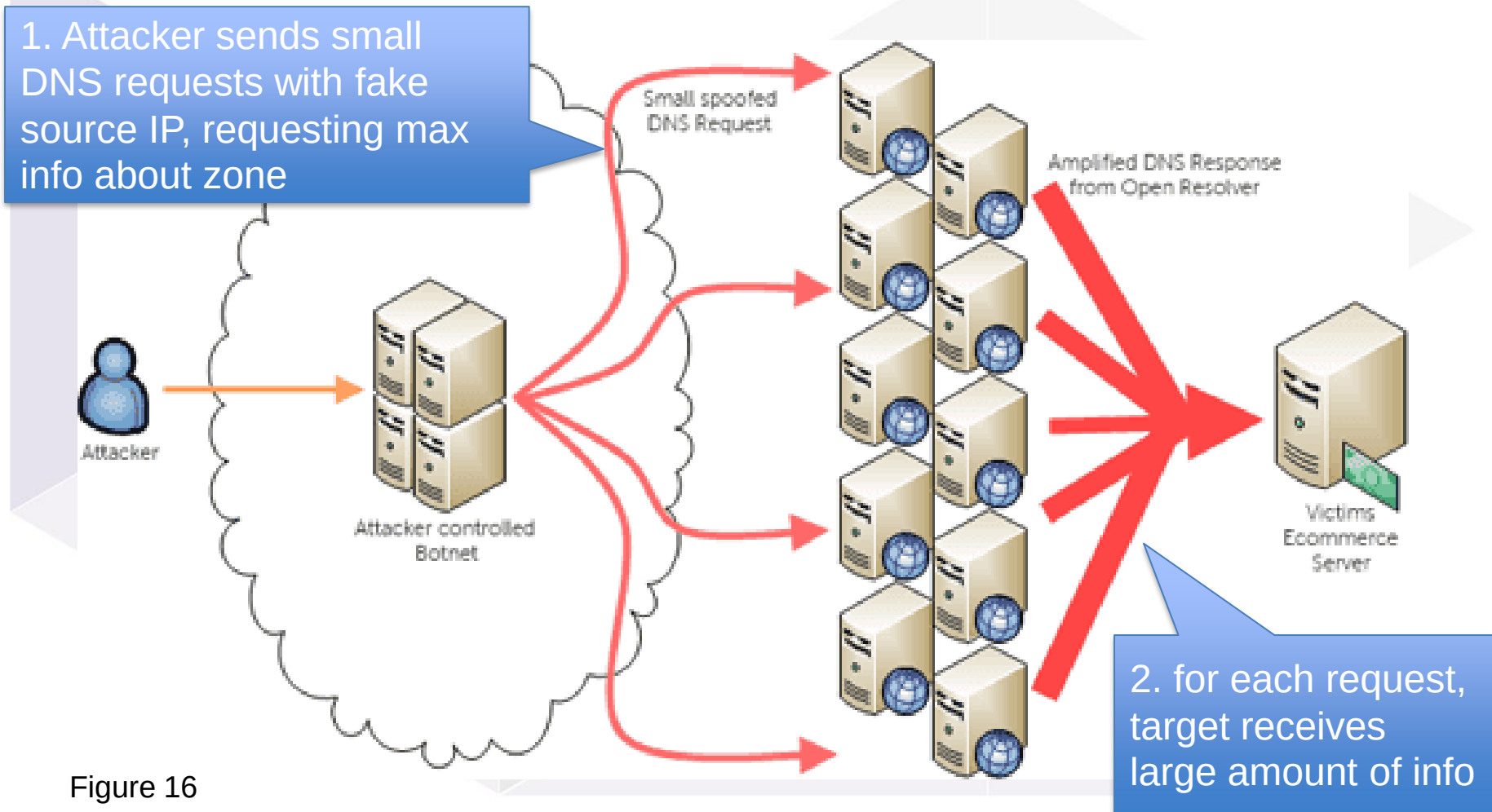


Figure 16

Cyberspies Hijacked the Internet Domains of Entire Countries

A mysterious new group called Sea Turtle targeted 40 organizations in a DNS hijacking spree.

Figure 17

DNS hijacking

- Attacker hacks into DNS server (hijacking)
- ...or intercepts DNS queries and responds with fake info (spoofing)
- You don't receive the real IP address, but that of a fake site.

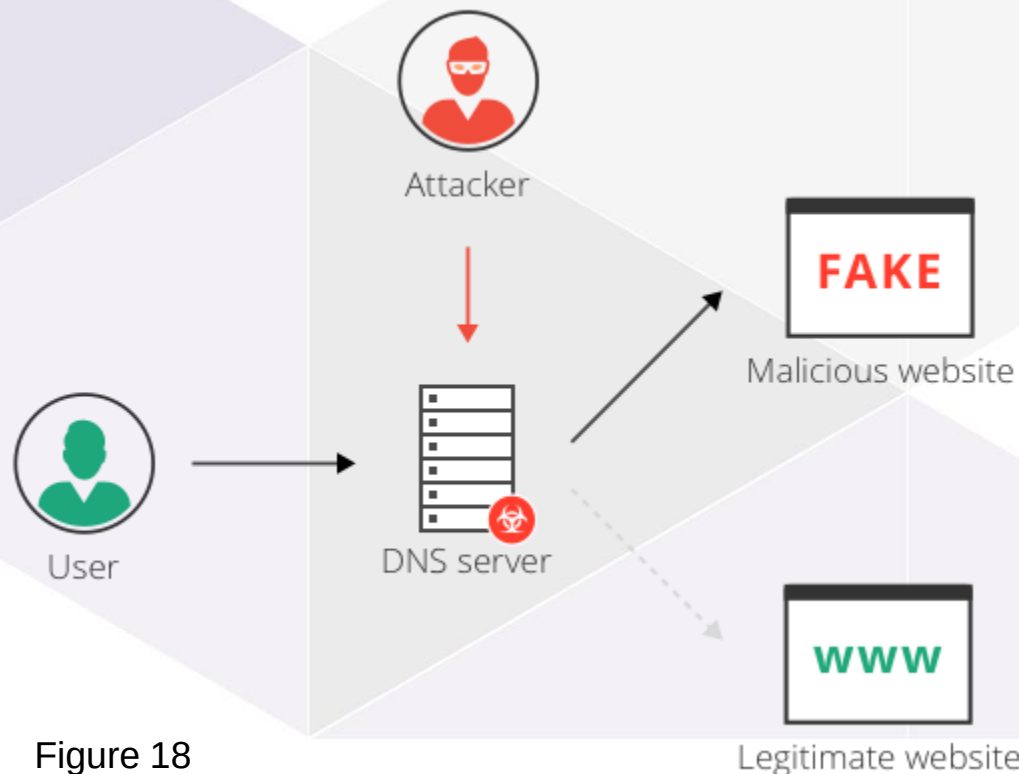


Figure 18

DNS over HTTPS or TLS

privacy

DNS is unencrypted:

- Anyone can listen in (provider, hacker, anyone)

solution?

DNS over HTTPS/SSL:

- Browsermakers (Google, Mozilla) are implementing this

but...

Instead of *anyone* having access your data, now it's gathered by one specific DNS provider.

DNSSec

- *DNSSec* signs DNS records with a digital signature
- Using public/private key cryptography
- This provides authentication (but not confidentiality)
- Similar to website certificates

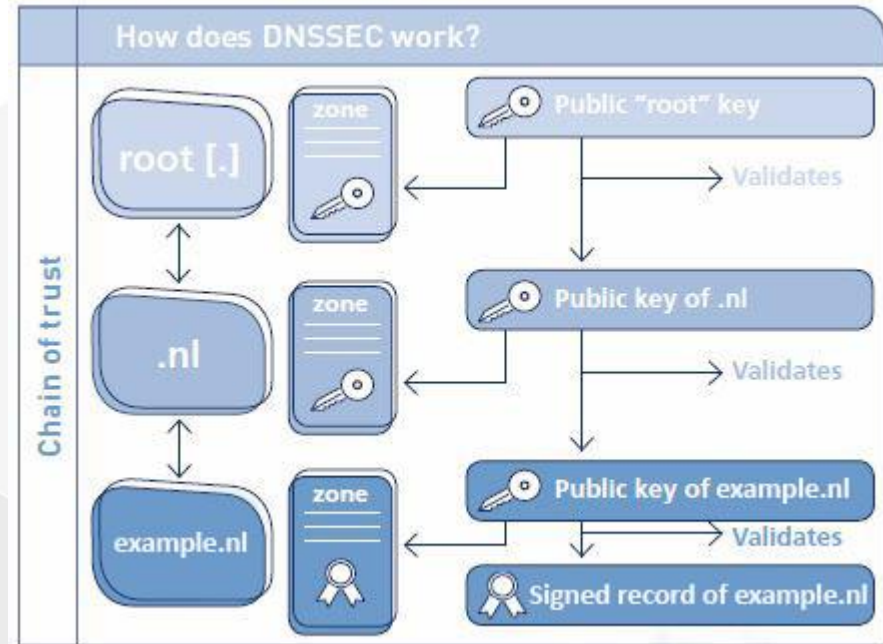


Figure 19

Agenda

- DNS
- **Routing**
- Routing Algorithms
- Routing Protocols

Routing

Reminder: routing & forwarding

Routing algorithms

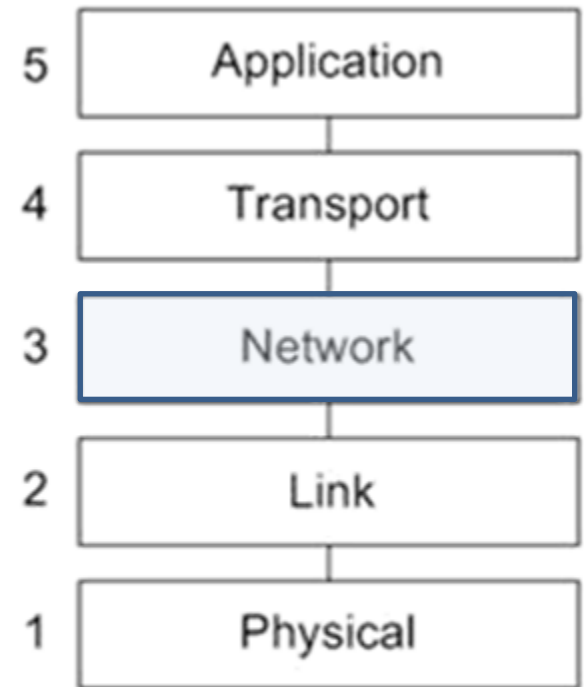
- Link State
- Distance Vector

Routing and Autonomous Systems

Routing Protocols

- Intra-AS Routing: OSPF
- Inter-AS Routing: BGP

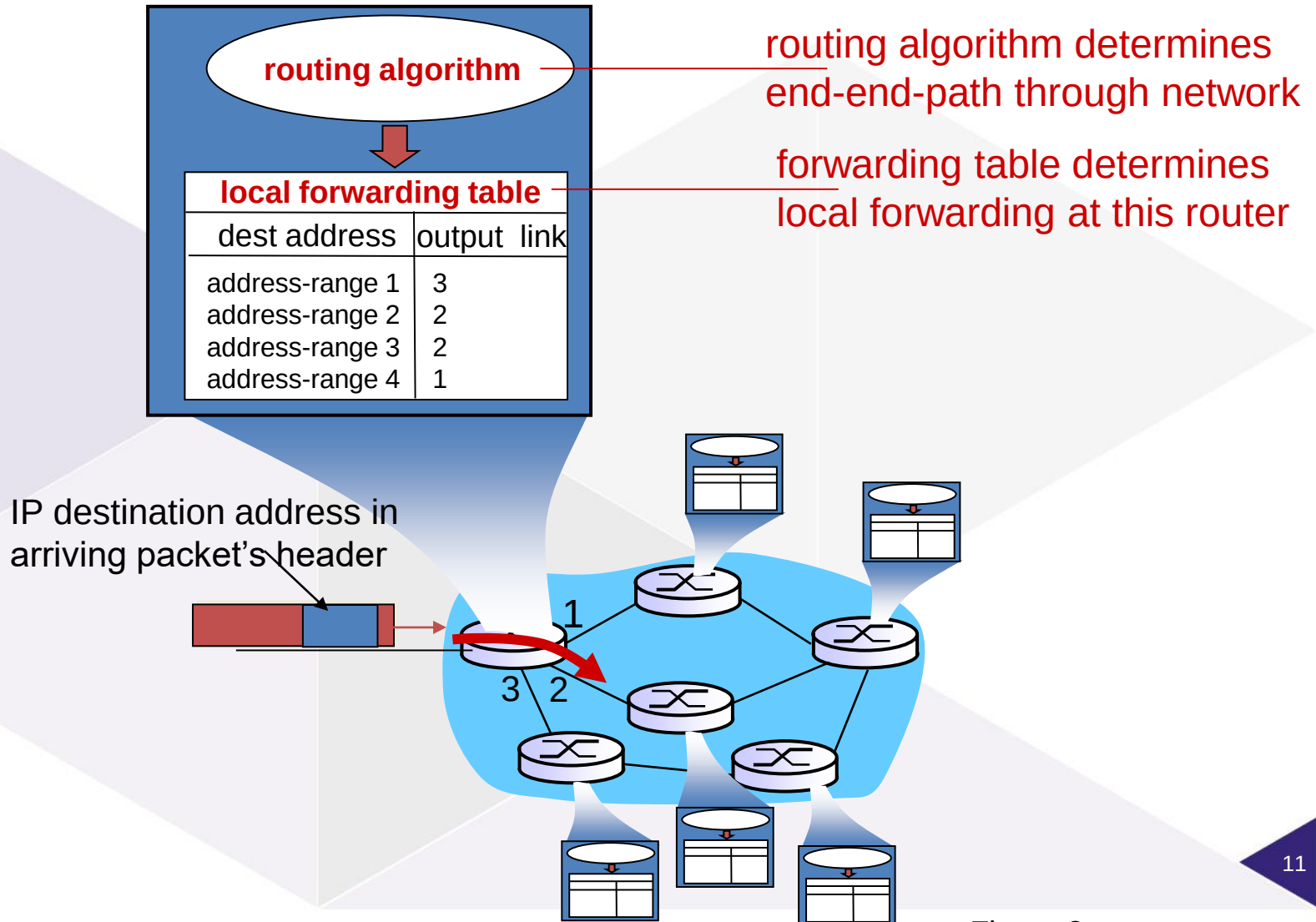
BGP hijacking



TCP/IP Model

Figure 1

Interplay between routing, forwarding



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- Routing Protocols

Link costs

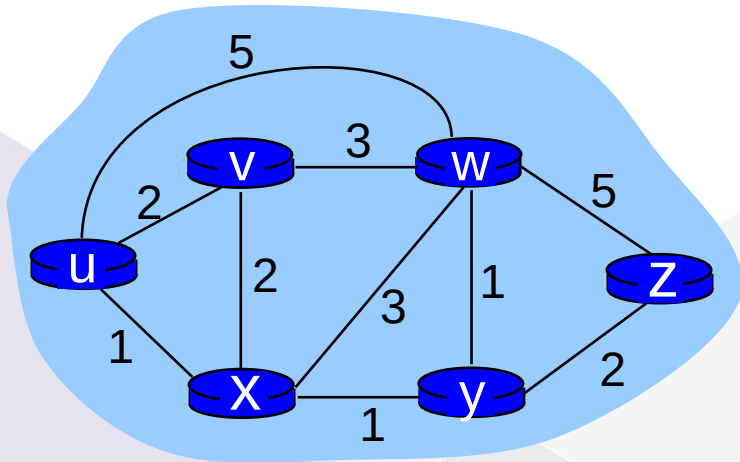


Figure 4

- For each link between two routers, we can set the link cost
- Cost could always be 1, or related to bandwidth, or related to congestion, or ...

$$\text{cost of path } (x_1, x_2, x_3, \dots, x_p) = c(x_1, x_2) + c(x_2, x_3) + \dots + c(x_{p-1}, x_p)$$

key question: what is the least-cost path between u and z ?

routing algorithm: algorithm that finds that least cost path

Routing algorithm classification

Q: global or decentralized information?

global:

- all routers have complete map (topology), link cost info

“link state” algorithms

decentralized:

- router knows physically-connected neighbors, link costs to neighbors
- iterative process of computation, exchange of info with neighbors

“distance vector” algorithms

A Link-State routing algorithm

Dijkstra's algorithm

net topology, link costs known to all nodes

- accomplished via “link state broadcast”
- all nodes have same info

computes least cost paths from one node (‘source’) to all other nodes

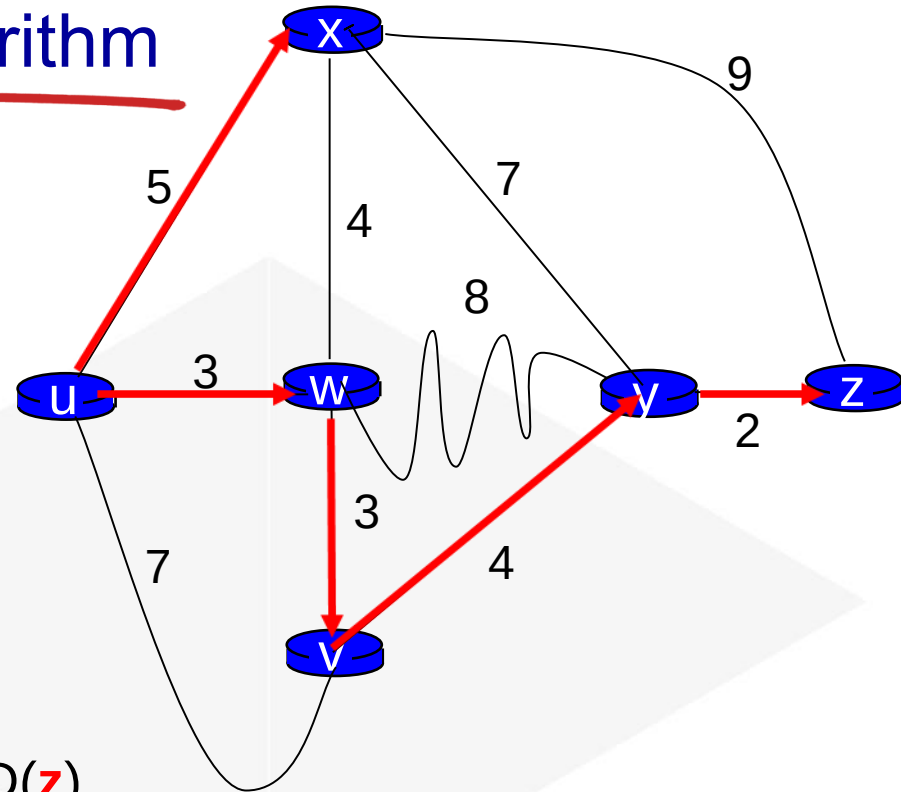
- Results in *forwarding table* for the source node

iterative: after k iterations, know least cost path to k destinations with certainty

Dijkstra's algorithm

Step 1

- ❖ Routers broadcast their link state
- ❖ Complete network topology now known to each router



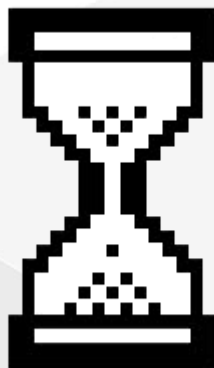
Step	N'	D(v) p(v)	D(w) p(w)	D(x) p(x)	D(y) p(y)	D(z) p(z)
0	u	7,u	3,u	5,u	∞	∞
1	uw	6,w		5,u	11,w	∞
2	uwx	6,w			11,w	14,x
3	uwxv				10,v	14,x
4	uwxvy					12,y
5	uwxvyz					

Step 2

- ❖ Using Dijkstra's algorithm, step by step...
- ❖ Each router calculates their own routing table with best routes to each destination

Figure 5

Break



Distance vector algorithm

iterative, asynchronous: each local iteration caused by:

local link cost change

DV update message from neighbor

distributed:

each node notifies neighbors *only* when its DV changes

- neighbors then notify their neighbors if necessary

decentralized:

router knows physically-connected neighbors, but only has estimates of best routes to other destinations

each node:

wait for (change in local link cost or msg from neighbor)

recompute estimates

if DV to any dest has changed, *notify* neighbors

Distance vector: example

In this example, the cost for each link is set to 1.
So, the total cost is simply the number of hops

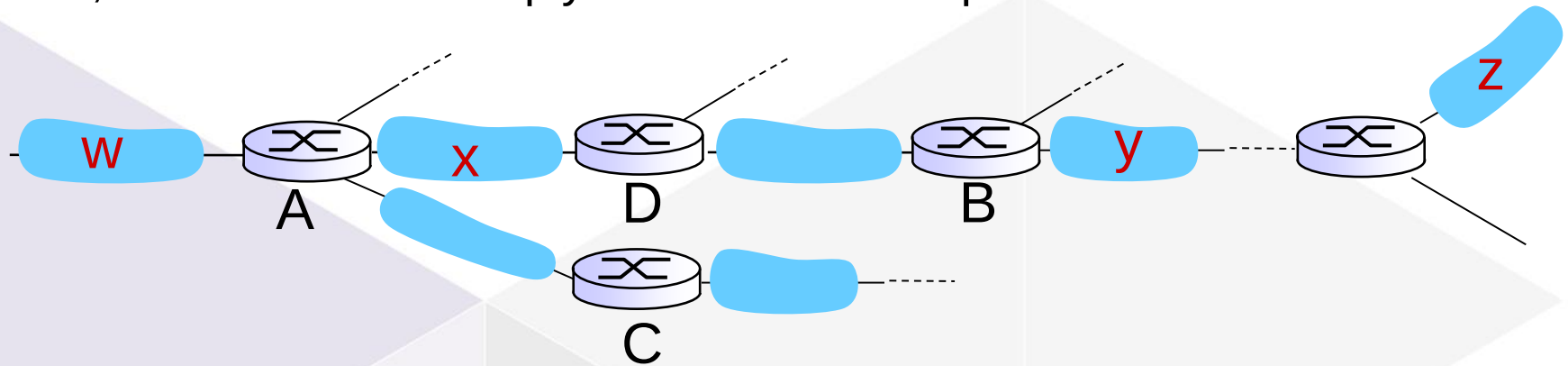


Figure 11

routing table in router D

destination subnet	next router	# hops to dest
W	A	2
y	B	2
Z	B	7
X	--	1
....		

Distance vector: example

A-to-D advertisement

dest	next	hops
W	-	1
X	-	1
Z	C	4
...		

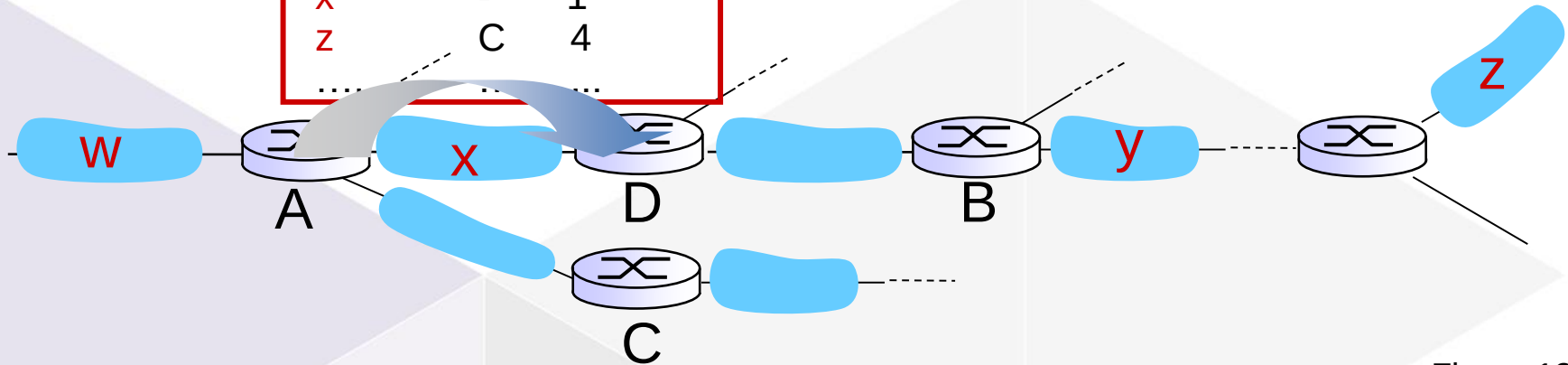


Figure 12

routing table in router D

destination subnet	next router	# hops to dest
W	A	2
Y	B	2
Z	B → A	7 → 5
X	--	1
....		

Agenda

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- **Routing Protocols**

Autonomous systems

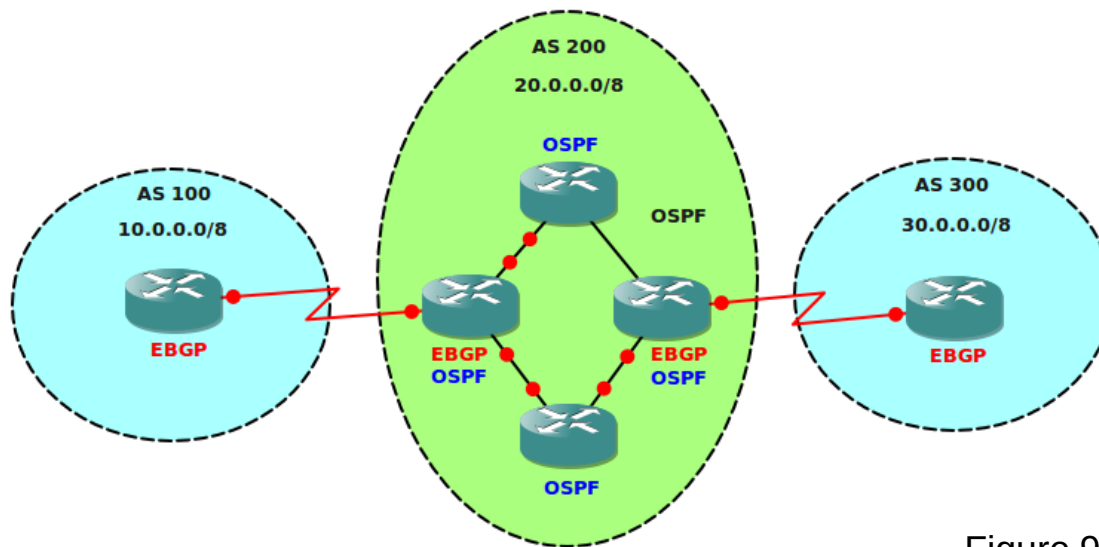


Figure 9

Autonomous system:

- A group of routers under the same control
- Belong to (for example) a single company or ISP

- Intra-AS routing: routing within AS (i.e. OSPF, RIP, IGRP)
- Inter-AS routing: routing between AS's (i.e. BGP)

Why different routing protocols?

policy:

inter-AS: admin wants control over how its traffic routed, who routes through its net.

intra-AS: single admin, so no policy decisions needed

scale:

hierarchical routing saves table size, reduced update traffic

performance:

intra-AS: can focus on performance

inter-AS: policy may dominate over performance

INTRA-AS ROUTING: OSPF (Open Shortest Path First)

“open”: publicly available
uses link state algorithm

OSPF advertisement router advertisements flooded to *entire* AS

- carried in OSPF messages directly over IP (rather than TCP or UDP)

Hierarchical OSPF in large domains.

security: all OSPF messages authenticated (to prevent malicious intrusion)

Internet inter-as routing: BGP

BGP (border gateway protocol): *the* de facto inter-domain routing protocol

- “glue that holds the Internet together”

BGP provides each AS a means to:

- **eBGP:** obtain subnet reachability information from neighboring ASes
- **iBGP:** propagate reachability information to all AS-internal routers.
- determine “good” routes to other networks based on reachability information and *policy*

allows subnet to advertise its existence to rest of Internet: *“I am here”*

EBGP, IBGP CONNECTIONS

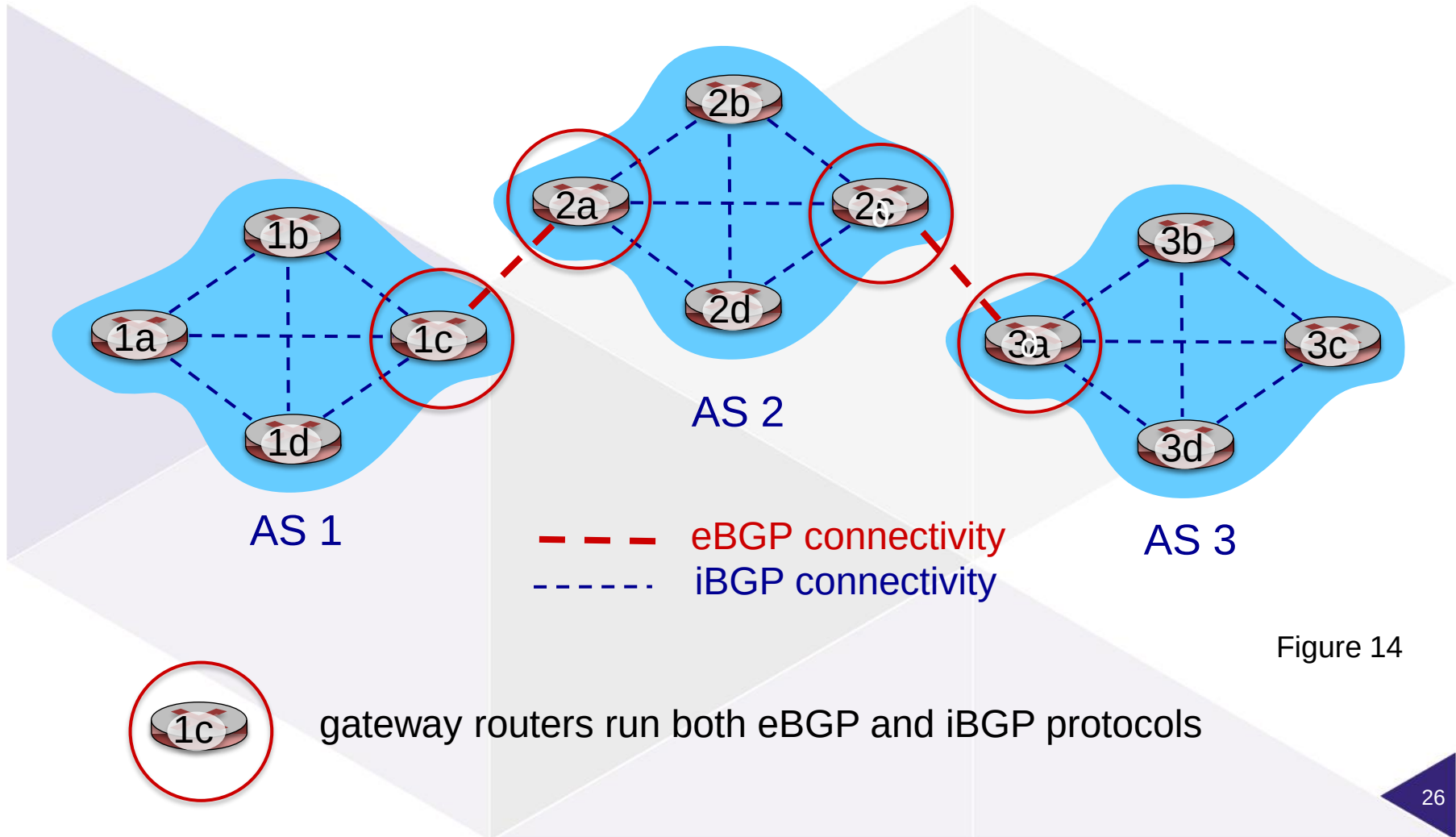


Figure 14

BGP basics

- **BGP session:** two BGP routers exchange BGP messages:
 - advertising *paths* to different destination network prefixes (BGP is a “path vector” protocol)

when AS3 gateway router 3a advertises path **AS3,X** to AS2 gateway router 2c:

- AS3 *promises* to AS2 it will forward datagrams towards X

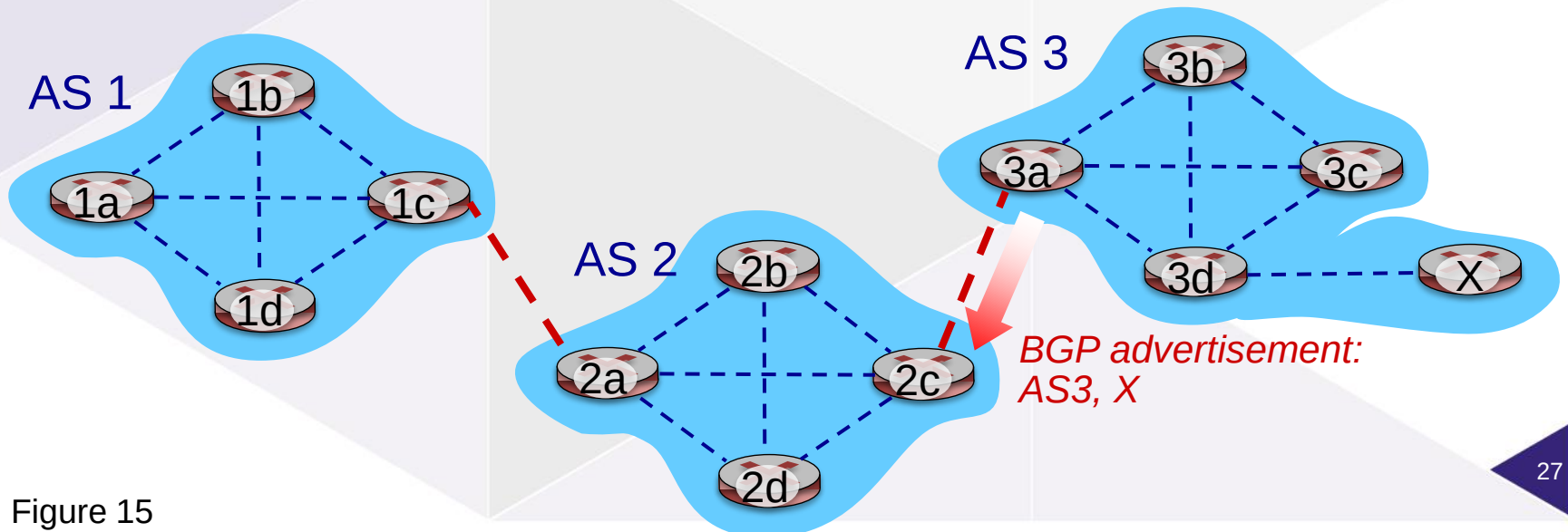


Figure 15

Path attributes and BGP routes

advertised network prefix includes BGP attributes:

- **AS-PATH**: list of ASes through which prefix advertisement has passed
- **NEXT-HOP**: indicates specific internal-AS router to next-hop AS

Policy-based routing:

- Policies determine whether a path is advertised and whether an advertised path will be accepted (e.g., never route through AS Y).

Route selection:

router may learn about more than one route to destination AS, selects route based on:

1. local preference value attribute: policy decision
2. shortest AS-PATH
3. closest NEXT-HOP router: hot potato routing
4. additional criteria

For two hours, a large chunk of European mobile traffic was rerouted through China

It was China Telecom, again. The same ISP accused last year of "hijacking the vital internet backbone of western countries."



By Catalin Cimpanu for Zero Day | June 7, 2019 -- 19:41 GMT (20:41 BST) | Topic: Security

Figure 20 - Simple mistake or Chinese state sponsored hackers?

BGP hijacking

- Can happen by mistake (configuration error by an ISP) or as an attack
- An AS announces it originates a prefix that it does not originate, or a shorter route
- Traffic will now be routed through this AS, where an attacker can intercept it
- Solution: validate announcements with certificate chains (*compare DNSSec, website certificates*)
- BGPSec met RPKI (Resource Public Key Infrastructure) can do this, but is not widely used yet

Sources

Figure 1,2,3,4,5,6,8,10,11,12,13,14,15: Kurose and K.W. Ross, Computer Networking, A Top-Down Approach, Pearson Education, 2013

Figure 9: Rene Molenaar, 26-5-2011, BGP for beginners, retrieved on 24-10-2017 from <https://gns3vault.com/blog/bgp-for-beginners/>

Figure 16: Peter, 16-10-2013, DNS amplification attacks, retrieved on 30-10-2019 from <https://blog.sflow.com/2013/10/dns-amplification-attacks.html>

Figure 17: Andy Greenberg, 17-4-2019. Cyberspies Hijacked the Internet Domains of Entire Countries, retrieved on 30-10-2019 from <https://www.wired.com/story/sea-turtle-dns-hijacking/>

Figure 18: Imperva. Domain name server (DNS) Hijacking, retrieved on 30-10-2019 from <https://www.imperva.com/learn/application-security/dns-hijacking-redirection/>

Figure 19:SIDN. DNSsec, retrieved on 30-10-2019 from <https://www.sidn.nl/veilig-internet/dnssec>

Figure 20:Catalin Cimpanu, 7-6-2019, For two hours, a large chunk of European mobile traffic was rerouted through China, retrieved on 30-10-2019 from <https://www.zdnet.com/article/for-two-hours-a-large-chunk-of-european-mobile-traffic-was-rerouted-through-china/>